

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All MMLU content is sourced from US standardized tests (GRE, USMLE), undergraduate course materials, and Oxford University Press books [Q19], with Professional Law supplemented by Harvard Law Library case law [Q77] and history items drawing on English primary sources [Q140, Q141]. The authorship team is entirely US-based [Q10] and funding is from US sources [Q87, Q88]. Empirical sampling confirms US-centric content saturation: US tort law [DATASET-D7], US contract law [DATASET-D8], US Social Security [DATASET-D9], US birth control statistics [DATASET-D10], US Populist movement [DATASET-D2], MCAT references [DATASET-D39], and US measurement units [DATASET-D29]. South Asian content is limited to three tangential references (India internet 2017, India vs. Congo university graduates, Jainism/samsara) and one colonially-framed Indian Uprising item [DATASET-D13, D40, D36, D18]. No content from BRRI, BARI, BAU, ICAR, PJTSAU, or ANGRAU was identified, and gap searches confirmed no Bengali- or Telugu-language agricultural NLP corpus exists in the public landscape [WEB-22, WEB-6]. The Mymensingh-dialect agricultural vocabulary, Telugu dry-land terminology, and cross-border water-policy content required for this deployment are absent by construction.

ID	Evidence
Q10	"Dan Hendrycks, UC Berkeley; Collin Burns, Columbia University; Steven Basart, UChicago; Andy Zou, UC Berkeley; Mantas Mazeika, UIUC; Dawn Song, UC Berkeley; Jacob Steinhardt, UC Berkeley" (p.1)
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q77	"We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)
Q87	"DH is supported by the NSF GRFP Fellowship and an Open Philanthropy Project Fellowship." (p.9)
Q88	"This research was also supported by the NSF Frontier Award 1804794." (p.9)
Q140	"This question refers to the following information." (p.19)
Q141	"English Parliament, Act of Supremacy, 1534" (p.19)
D2	"The song, and the movement that it was connected to, highlight which of the following developments in the broader society in the late 1800s?" (re: "The Hayseed", US Populist movement) — US-specific historical content with no South Asian relevance
D7	A truck driver was assigned to drive a huge tractor-trailer loaded with logs. The logs were being sold by a lumber company... the girl was struck by the tractor-trailer and was seriously injured. — US tort law scenario — US legal context
D8	An avid baseball fan learned that a local professional baseball club was conducting a baseball camp... forwarded the club a \$50 registration deposit — US contract law — US sports/cultural context
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D13	As of 2017, what fraction of the population in **India** used the internet in the past three months? (A) 11% (B) 26% (C) 41% (D) 56% — South Asia-referenced factual question, but static 2017 statistic

D18	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters... British journalist William Howard Russell, My Indian Mutiny Diary, 1860 — British colonial framing of Indian history
D29	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? — US units (miles per hour) — not metric; minor cultural assumption
D36	In Jainism, what is the cycle that one must liberate oneself from? (A) Punya (B) Samsara — South Asian religion reference (Jainism) — one of very few South Asian cultural anchors in the dataset
D39	When preparing for the **MCAT exam** , a student begins studying electrochemical cells. — MCAT (US Medical College Admissions Test) reference — US academic system assumed
D40	At the time of independence, there were already hundreds of thousands of university graduates in **India** , but hardly any at all in **Congo** . — India referenced but as a factual datapoint about independence-era education, not agro-ecological
WEB-6	ICICC 2024 Telugu agricultural perplexity study explicitly notes lack of Telugu agricultural NLP resources (https://doi.org/10.1007/978-981-97-4152-6_38)
WEB-22	AI4Bharat IndicNLP Catalog — no Bangladeshi-Bengali or Mymensingh-dialect agricultural corpus listed (https://ai4bharat.github.io/indicnlp_catalog/)

Strengths

- A handful of global_facts and world_religions items reference South Asia, providing a minimal foothold of non-exclusively-Western reference [DATASET-D13, D36, D40]
- PDF/separate-page sourcing reduces verbatim memorization risk, preserving construct measurement integrity for the topics it does cover [Q110]

ID	Evidence
Q110	"We also note that most of our questions came from PDFs or websites where questions and answers are on separate pages." (p.14)
D13	As of 2017, what fraction of the population in **India** used the internet in the past three months? (A) 11% (B) 26% (C) 41% (D) 56% — South Asia-referenced factual question, but static 2017 statistic
D36	In Jainism, what is the cycle that one must liberate oneself from? (A) Punya (B) Samsara — South Asian religion reference (Jainism) — one of very few South Asian cultural anchors in the dataset
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Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — many MMLU items require US-specific cultural and institutional knowledge (US tort law, Social Security, MCAT system, US demographic statistics) [DATASET-D7, D9, D10, D11, D39]. Conversely, items requiring Mymensingh, Bangladeshi, or Telugu-region knowledge are absent.

IC-2: Culturally sensitive content (moral_scenarios anchored in 'US 2020 norms', US-centric birth control and aging facts) does not align with target deployment culture [DATASET-D1, D10, D11, D12]. Cross-border Bangladesh-India water-sharing topics — actively contested as of 2025 [WEB-1, WEB-2, WEB-4] — are not represented at all.

IC-3: Inputs requiring Western-specific knowledge are pervasive: US Constitutional structure [DATASET-D3, D5], US legislative history [DATASET-D4], Burgess Chicago-school urban model [DATASET-D20], UK educational sociology [DATASET-D26]. These do not transfer to South Asian agricultural contexts.

IC-4: INSUFFICIENT DOCUMENTATION on whether regional annotators reviewed any items; paper documents only US graduate/undergraduate collectors [Q19] and US MTurk baseline [Q22]. Full assessment would require recruiting Mymensingh, Telugu, and West Bengal agricultural experts to flag culturally sensitive instances.

IC-5: Combined with Q19's exclusively US sourcing and the empirical pattern across 175+ sampled examples, content validity for this deployment is severely compromised.

ID	Evidence
Q19	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q22	"Human-level accuracy on this test varies. Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test. Meanwhile, expert-level performance can be far higher. For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task. If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
Q81	"Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations." (p.8)
D1	For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, **according to ordinary moral standards in the US as of 2020**? — Question explicitly frames moral standards as US-specific, 2020 vintage
D3	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — Entirely US constitutional law content
D4	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president. — US-specific legislative history
D5	In American government, the power to declare war rests with Congress. — US government structure; no relevance to South Asian governance or environmental policy
D7	A truck driver was assigned to drive a huge tractor-trailer loaded with logs. The logs were being sold by a lumber company... the girl was struck by the tractor-trailer and was seriously injured. — US tort law scenario — US legal context
D9	The age at which one can receive full benefits from social security" with answer "Has been increasing — US Social Security system — entirely US institutional context
D10	Which of the following is the most common method of birth control for married couples in the **U.S.** — Explicitly US demographic data
D11	From 1988 to 1990 among heterosexuals in the **US**, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US-specific behavioral survey data
D12	"For which of these two scenarios does the main character ... do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?" (repeated framing across all 5 examples) — Every moral_scenarios question frames correctness relative to US 2020 norms
D20	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric **zone model of urban form**? — US urban planning model (Burgess model, Chicago School)
D26	Smith & Tomlinson argued that school character far outweighed ethnic background in determining educational success — UK-centric sociological research
D39	When preparing for the **MCAT exam**, a student begins studying electrochemical cells. — MCAT (US Medical College Admissions Test) reference — US academic system assumed
WEB-1	Active Bangladesh-India treaty renewal context — divergent correct answers likely (https://www.tbsnews.net/foreign-policy/bangladeshi-indian-engineers-monitor-water-flows-under-ganges-water-sharing-treaty)
WEB-2	Teesta dispute unresolved — Bangladesh and India would adjudicate water-allocation questions differently (https://en.wikipedia.org/wiki/Teesta_Water_Dispute)
WEB-4	Teesta Master Plan with Chinese financing — Bangladeshi vs Indian framings differ (https://www.tbsnews.net/features/panorama/teesta-master-plan-and-longstanding-bangladesh-india-water-politics-1072696)
WEB-6	ICICC 2024 Telugu agricultural perplexity study explicitly notes lack of Telugu agricultural NLP resources (https://doi.org/10.1007/978-981-97-4152-6_38)

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Information Gaps

- Whether any unsampled config contains incidental Bangladesh- or Telugu-region agricultural content
- Whether annotators included any with South Asian regional expertise (paper does not document annotator demographics beyond US institutional affiliation)

Requires Expert Verification

- BRRRI/BARI/BAU agricultural extension experts could confirm absence of dialect-specific Mymensingh terminology
- Telugu agricultural linguistic experts (PJ TSAU/ANGRAU) could confirm absence of Deccan dry-land cropping vocabulary

Remediation

Gap 1: Content is entirely sourced from US/Anglophone institutions [Q19, Q77]; no Bengali- or Telugu-language agricultural content; cross-border policy context (Ganges/Teesta) absent.

Recommendation: Author a parallel evaluation set sourcing items from BRRRI/BARI technical bulletins, PJ TSAU/ANGRAU extension publications, and Joint Rivers Commission documents. Use the AgriPriceBD [WEB-23] LLM-assisted PDF parsing methodology to digitize Bengali agricultural reports. Build dialect-aware Mymensingh-Bengali corpus through BAU faculty elicitation (acknowledging this is unsearchable web-research territory).

ID	Evidence
Q19	“The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books.” (p.3)
Q77	“We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard’s Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy.” (p.8)
WEB-23	https://arxiv.org/abs/2604.06227